

2023



Democratizing Access to AWS Cost Optimization with Canva

Learn how Canva reduces costs using strategies adopted by 100 million monthly active users.



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A white icon of a text selection tool (blue 'E' and orange 'I') on a dark background.

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2

46% reduction

in compute costs in less than 2 years

Scaled from zero

to accomodate 160 million monthly active users

Democratizes cost optimization

information across product teams

Accelerates

development of new features

Overview

The global visual communication company [Canva](#) was looking to optimize compute costs after a period of exceptional growth. The infrastructure team also wanted to empower the engineering teams with greater insights into cost. That way, development teams wouldn't have to rely on the central infrastructure team to see how their compute choices affected monthly average users or other parameters that might drive even better decision-making.

From its inception, Canva has built its platform on Amazon Web Services (AWS). Working alongside AWS teams, Canva's infrastructure team identified and implemented a combination of AWS purchase models to reduce compute costs by 46 percent in less than 2 years. The company used AWS cost optimization tools to scale in a way that would cost-effectively accommodate growth, and engineering teams used the insights they gained to accelerate the development of innovative new features for customers.



Opportunity | Using the Breadth of Amazon EC2 Purchase Models and Cost Optimization Tools to Control Costs While Scaling to 160 Million Monthly Active Users for Canva

Launched in Australia in early 2013, Canva makes professional-quality design accessible for more than 160 million monthly active users who have created 15 billion designs across 190 countries. To meet compute needs that fluctuate up to 200 designs per second, Canva uses [Amazon Elastic Compute Cloud](#) (Amazon EC2), which offers secure and resizable compute capacity for virtually any workload. “Thanks to our impressive growth, healthy revenues, and fortunate positioning, cost hasn’t been a primary concern for us,” says Fawaz Ahmed, Canva’s director of engineering. “However, macroeconomic shifts have led to a shift in focus toward optimization and making sure that every dollar is spent wisely.”

The company experiences significant spikes in usage with the introduction of popular features. For example, users have created more than one billion videos since Canva launched a video editing suite in 2021. Likewise, the company gained users when it launched Magic Write, an innovative text generator powered by artificial intelligence. Canva also offers an increasing number of ways for users to download their designs, such as PDFs, PNGs, PowerPoints, and other options. Canva wanted to continue the momentum of developing these compute-heavy features while optimizing costs.

Solution | Cutting Compute Costs by 46 Percent Using AWS Cost Optimization Tools

An [AWS Partner](#) since August 2019 and customer since 2013, Canva holds monthly business reviews with its AWS technical account management team, which analyzes Canva’s compute usage and proactively provides the best options for cost optimization. “It’s very much a joint effort,” says Ahmed. “We work very closely together, and the AWS team is always there to jump in with advice and recommendations.”

For example, Canva runs its video and streaming functions as containerized workloads on [Amazon Elastic Container Service](#) (Amazon ECS), a fully managed container orchestration service that simplifies deployment, management, and scaling of containerized applications. These Amazon ECS workloads use [Amazon ECS cluster autoscaling](#), a feature that automatically manages the scaling of Amazon EC2 instances registered to the cluster. Canva uses a dedicated capacity provider that runs on [Amazon EC2 Spot Instances](#). Spot Instances help organizations take



advantage of unused Amazon EC2 capacity and are available at up to a 90 percent discount compared with [Amazon EC2 On-Demand Pricing](#).

In early 2022, Canva put together a financial operations team that includes personnel from finance, procurement, and engineering. The team sought to democratize information about cost optimization among product teams. “Saying that a compute cluster costs X dollars isn’t as useful as saying that it contributes to a certain number of monthly active users or a certain strategy,” says Ahmed. “Cost is just one dimension. When we give meaning to costs, that empowers our product teams to make better decisions and more informed choices for their workloads and users.”

Canva uses AWS tools to gain greater visibility into cost management so that teams can match instance type to workload. For example, it implements [Cloud Financial Management](#) with AWS, which organizations can use to transform their businesses with cost transparency, control, forecasting, and optimization. Also, some account teams use [AWS Compute Optimizer](#), which provides recommendations to optimize the use of AWS resources. “We want to focus on how we can grow more purposefully through conscious business decisions,” says Ahmed. “Growth doesn’t necessarily mean that you’re continually going to have a bigger and bigger bill. If you optimize, growth shouldn’t directly correlate with cost.”

After successfully using Spot Instances to optimize costs for containerized workloads, Canva has begun to use them for higher-priority jobs. It maintains fallback instances using [Amazon EC2 Reserved Instances](#), which provide a discount of up to 72 percent compared with On-Demand pricing and provide a capacity reservation when used in a specific Availability Zone. Additionally, Canva distinguishes the type of compute instance by the tier of the user. For example, the projects created by Canva Free users, who are on a free plan, run on Spot Instances. By contrast, the projects of Canva Pro users are backed by Amazon EC2 On-Demand Instances for greater reliability as part of Canva’s [Savings Plans](#), a flexible pricing model that organizations can use to reduce their bills by up to 72 percent compared with On-Demand Pricing, in exchange for a 1- or 3-year hourly spend commitment. All told, Canva has reduced compute costs by 46 percent through a finely tuned combination of purchase options on AWS.

Outcome | Reinvesting in Artificial Intelligence–Driven Products While Optimizing Costs on AWS

For additional cost savings, Canva has begun to run some of its OpenSearch workloads using instances powered by [AWS Graviton processors](#), which are designed by AWS to deliver the best price performance for cloud workloads running in Amazon EC2. By 2026, Canva will be looking at powering additional workloads using AWS Graviton–based instances as a next wave of its compute cost-optimization strategy. Additionally, AWS Graviton-based instances use up to 60 percent less energy than comparable Amazon EC2 instances, contributing to the reduction of the carbon footprint.

As the company grows, Canva engineers use AWS to off-load scaling and other manually intensive computing tasks so that they can focus on getting new products to market.

The company built a [text-to-image](#) feature in 3 weeks, and it continues to introduce features that capitalize on machine learning (ML) and artificial intelligence. “In terms of ML, we’re able to focus a lot more on the training and development of models,” says Ahmed. “We provide them for customers and get to market faster than we otherwise would have. AWS provides the tools and the teamwork for us to cost-effectively implement the compute offerings that work best for us.”



About Canva

Canva offers visual communication software to 160 million monthly active users across 190 countries. Its offerings include the Visual Suite, a suite of workplace products, and Magic Studio, a collection of AI-powered tools.

AWS Services Used

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

[Learn more »](#)

Amazon EC2 Spot Instances

Amazon EC2 Spot Instances let you take advantage of unused EC2 capacity in the AWS cloud and are available at up to a 90% discount compared to On-Demand prices.

[Learn more »](#)

Amazon EC2 Reserved Instances

Amazon EC2 Reserved Instances (RI) provide a significant discount (up to 72%) compared to On-Demand pricing and provide a capacity reservation when used in a specific Availability Zone.

[Learn more »](#)

AWS Graviton Processors

AWS Graviton is a family of processors designed to deliver the best price performance for workloads running in Amazon Elastic Compute Cloud (Amazon EC2).

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